



SPIRIT 2013 Checklist: Recommended items to address in a clinical trial protocol and related documents*

Section/item	Item No	Description	Addressed on page number
Administrative information			
Title	1	Descriptive title identifying the study design, population, interventions, and, if applicable, trial acronym	1
Trial registration	2a	Trial identifier and registry name. If not yet registered, name of intended registry	4
	2b	All items from the World Health Organization Trial Registration Data Set	None
Protocol version	3	Date and version identifier	4
Funding	4	Sources and types of financial, material, and other support	15
Roles and responsibilities	5a	Names, affiliations, and roles of protocol contributors	14
	5b	Name and contact information for the trial sponsor	1
	5c	Role of study sponsor and funders, if any, in study design; collection, management, analysis, and interpretation of data; writing of the report; and the decision to submit the report for publication, including whether they will have ultimate authority over any of these activities	13

5d	Composition, roles, and responsibilities of the coordinating centre, steering committee, endpoint adjudication committee, data management team, and other individuals or groups overseeing the trial, if applicable (see Item 21a for data monitoring committee)	13
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Introduction

Background and rationale	6a	Description of research question and justification for undertaking the trial, including summary of relevant studies (published and unpublished) examining benefits and harms for each intervention	2-4
	6b	Explanation for choice of comparators	3-4
Objectives	7	Specific objectives or hypotheses	4
Trial design	8	Description of trial design including type of trial (eg, parallel group, crossover, factorial, single group), allocation ratio, and framework (eg, superiority, equivalence, noninferiority, exploratory)	4

Methods: Participants, interventions, and outcomes

Study setting	9	Description of study settings (eg, community clinic, academic hospital) and list of countries where data will be collected. Reference to where list of study sites can be obtained	4
Eligibility criteria	10	Inclusion and exclusion criteria for participants. If applicable, eligibility criteria for study centres and individuals who will perform the interventions (eg, surgeons, psychotherapists)	5-6
Interventions	11a	Interventions for each group with sufficient detail to allow replication, including how and when they will be administered	6-7
	11b	Criteria for discontinuing or modifying allocated interventions for a given trial participant (eg, drug dose change in response to harms, participant request, or improving/worsening disease)	14
	11c	Strategies to improve adherence to intervention protocols, and any procedures for monitoring adherence (eg, drug tablet return, laboratory tests)	None
	11d	Relevant concomitant care and interventions that are permitted or prohibited during the trial	None

Outcomes	12	Primary, secondary, and other outcomes, including the specific measurement variable (eg, systolic blood pressure), analysis metric (eg, change from baseline, final value, time to event), method of aggregation (eg, median, proportion), and time point for each outcome. Explanation of the clinical relevance of chosen efficacy and harm outcomes is strongly recommended	7-11
Participant timeline	13	Time schedule of enrolment, interventions (including any run-ins and washouts), assessments, and visits for participants. A schematic diagram is highly recommended (see Figure)	4
Sample size	14	Estimated number of participants needed to achieve study objectives and how it was determined, including clinical and statistical assumptions supporting any sample size calculations	5
Recruitment	15	Strategies for achieving adequate participant enrolment to reach target sample size	4-5

Methods: Assignment of interventions (for controlled trials)

Allocation:

Sequence generation	16a	Method of generating the allocation sequence (eg, computer-generated random numbers), and list of any factors for stratification. To reduce predictability of a random sequence, details of any planned restriction (eg, blocking) should be provided in a separate document that is unavailable to those who enrol participants or assign interventions	6
Allocation concealment mechanism	16b	Mechanism of implementing the allocation sequence (eg, central telephone; sequentially numbered, opaque, sealed envelopes), describing any steps to conceal the sequence until interventions are assigned	6
Implementation	16c	Who will generate the allocation sequence, who will enrol participants, and who will assign participants to interventions	None
Blinding (masking)	17a	Who will be blinded after assignment to interventions (eg, trial participants, care providers, outcome assessors, data analysts), and how	6
	17b	If blinded, circumstances under which unblinding is permissible, and procedure for revealing a participant's allocated intervention during the trial	None

Methods: Data collection, management, and analysis

Data collection methods	18a	Plans for assessment and collection of outcome, baseline, and other trial data, including any related processes to promote data quality (eg, duplicate measurements, training of assessors) and a description of study instruments (eg, questionnaires, laboratory tests) along with their reliability and validity, if known. Reference to where data collection forms can be found, if not in the protocol	13
	18b	Plans to promote participant retention and complete follow-up, including list of any outcome data to be collected for participants who discontinue or deviate from intervention protocols	None
Data management	19	Plans for data entry, coding, security, and storage, including any related processes to promote data quality (eg, double data entry; range checks for data values). Reference to where details of data management procedures can be found, if not in the protocol	13
Statistical methods	20a	Statistical methods for analysing primary and secondary outcomes. Reference to where other details of the statistical analysis plan can be found, if not in the protocol	12
	20b	Methods for any additional analyses (eg, subgroup and adjusted analyses)	None
	20c	Definition of analysis population relating to protocol non-adherence (eg, as randomised analysis), and any statistical methods to handle missing data (eg, multiple imputation)	None
Methods: Monitoring			
Data monitoring	21a	Composition of data monitoring committee (DMC); summary of its role and reporting structure; statement of whether it is independent from the sponsor and competing interests; and reference to where further details about its charter can be found, if not in the protocol. Alternatively, an explanation of why a DMC is not needed	12
	21b	Description of any interim analyses and stopping guidelines, including who will have access to these interim results and make the final decision to terminate the trial	11-12
Harms	22	Plans for collecting, assessing, reporting, and managing solicited and spontaneously reported adverse events and other unintended effects of trial interventions or trial conduct	14
Auditing	23	Frequency and procedures for auditing trial conduct, if any, and whether the process will be independent from investigators and the sponsor	None

Ethics and dissemination

Research ethics approval	24	Plans for seeking research ethics committee/institutional review board (REC/IRB) approval	4
Protocol amendments	25	Plans for communicating important protocol modifications (eg, changes to eligibility criteria, outcomes, analyses) to relevant parties (eg, investigators, REC/IRBs, trial participants, trial registries, journals, regulators)	14
Consent or assent	26a	Who will obtain informed consent or assent from potential trial participants or authorised surrogates, and how (see Item 32)	4-5
	26b	Additional consent provisions for collection and use of participant data and biological specimens in ancillary studies, if applicable	None
Confidentiality	27	How personal information about potential and enrolled participants will be collected, shared, and maintained in order to protect confidentiality before, during, and after the trial	13
Declaration of interests	28	Financial and other competing interests for principal investigators for the overall trial and each study site	15
Access to data	29	Statement of who will have access to the final trial dataset, and disclosure of contractual agreements that limit such access for investigators	13
Ancillary and post-trial care	30	Provisions, if any, for ancillary and post-trial care, and for compensation to those who suffer harm from trial participation	None
Dissemination policy	31a	Plans for investigators and sponsor to communicate trial results to participants, healthcare professionals, the public, and other relevant groups (eg, via publication, reporting in results databases, or other data sharing arrangements), including any publication restrictions	14
	31b	Authorship eligibility guidelines and any intended use of professional writers	None
	31c	Plans, if any, for granting public access to the full protocol, participant-level dataset, and statistical code	15
Appendices			
Informed consent materials	32	Model consent form and other related documentation given to participants and authorised surrogates	4-5

Biological specimens	33	Plans for collection, laboratory evaluation, and storage of biological specimens for genetic or molecular analysis in the current trial and for future use in ancillary studies, if applicable	None
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*It is strongly recommended that this checklist be read in conjunction with the SPIRIT 2013 Explanation & Elaboration for important clarification on the items. Amendments to the protocol should be tracked and dated. The SPIRIT checklist is copyrighted by the SPIRIT Group under the Creative Commons [“Attribution-NonCommercial-NoDerivs 3.0 Unported”](#) license.

Informed Consent Form for Subjects

Hello, welcome to the experimental study of "The Effects of Tai Chi on Insomnia Based on Multi-modal Brain Functional Connection and Signal Extraction", which will be completed in the Institute of Sports Medicine and Health at Chengdu Sport University, Affiliated Sport Hospital of Chengdu Sport University. 12 weeks of Tai Chi, brain function feature extraction, and sleep quality index testing will be conducted. The subjects will undergo 12 weeks of Tai Chi, brain function feature extraction, and sleep quality index test.

(1) Purpose and content of the experiment

We will use relevant tests such as Resting-State Functional Magnetic Resonance Imaging (rs-fMRI), Functional Near Infrared Tester (fNIRS), and Sleep Related Index, the results of which will help to assess functional brain connectivity, cerebral oxygen signaling, and quality of sleep to facilitate comparisons of pre- and post-intervention effects. You can ask any questions you may have about the testing procedure and results, and please consult us if you have concerns or questions.

(2) Training Instructions

Subjects should drink normal water and avoid eating for 3 hours before training. In addition, subjects should try to keep their diet as normal as possible during the training intervention without additional interventions. Thank you very much for your cooperation.

(3) Possible risks and discomforts

Since a single intervention lasted up to 60 minutes and most of the subjects had not been exposed to non-pharmacological therapies, discomfort due to nervousness and unforeseen situations such as abnormal blood pressure, dizziness, chest tightness, and panic may occur during testing and training.

(4) Responsibilities and obligations of participants

Participants should report any discomfort, such as nausea, pain in the joints, difficulty in breathing, etc., during training and testing. When you feel particularly fatigued or unwell, you should immediately ask the relevant staff to stop training or testing.

(5) Voluntary participation

Please make sure you are in good health to participate in this training experiment, and if you decide to participate in the experiment, we promise to keep all of your personal information confidential.

Subject informed consent:

I have been informed of the research process, purpose, and significance of the experiment, and I am clear about who I should approach to solve problems encountered in the experiment, I agree to participate in this experiment and follow the requirements strictly.

Subject's signature: phone number: time:

If you have any questions about the experiment, please contact: Jinying Li, contact number: 178XXXX0623

受试者知情同意书

您好，欢迎参加“基于多模态脑功能连接及信号提取的太极拳干预失眠症的效果研究”的实验研究，本研究将在成都体育学院运动医学与健康研究所和成都体育学院附属体育医院完成，受试者将进行 12 周的太极拳训练、脑功能特征提取和睡眠质量指标测试。

（1）实验目的及内容

我们将使用静息态功能磁共振成像（rs-fMRI）、功能性近红外测试仪（fNIRS）和睡眠相关指数等相关测试，测试的结果有助于评估脑功能连接、脑氧信号及睡眠质量，便于干预前后效果的对比。您可以提出任何有关测试步骤和结果的问题，如果有顾虑或疑问，请咨询我们。

（2）训练须知

受试者在训练前需正常饮水、并在训练前的 3 小时应避免进食。此外，受试者在训练干预期间饮食应尽量保持正常，不进行额外的干预。非常感谢您的配合。

（3）可能出现的风险和不适

由于单次干预时间长达 60 分钟，且受试者大多没有接触过非药物疗法，因此测试和训练过程中可能出现因紧张心理引起的不适感，以及出现血压异常、头晕、胸闷、心慌等意外情况。

（4）参与者的责任和义务

参与者在训练及测试过程中出现任何不适现象，如：恶心、关节处疼痛、呼吸困难等应及时报告。当您感到特别疲劳或不适时，应立即与相关工作人员要求停止训练或测试。

（5）自愿参加

请在确保自己身体健康的情况下参与本次训练实验，如果您决定参加实验，我们承诺对您的一切个人资料进行保密。

受试者知情同意书：

我已知晓实验的研究过程、目的和意义，并明确在实验中遇到问题应找谁解决，我同意参加本实验，严格按照要求进行。

受试者签名：

电话：

时间：

如您对实验有任何疑问，请联系：李津莹，联系电话：178XXXX0623